

VERSION: 1-1

UNIVERSITY OF TORONTO

FACULTY OF APPLIED SCIENCE AND ENGINEERING

Multiple Choice Assessment

August 25 - 26, 2021

First Year

APS164 - INTRODUCTORY CHEMISTRY FROM A MATERIALS
PERSPECTIVE

Exam Type: X (Duration 2.5 hours, plus 30 minutes for submission, no extensions. Open Book; Internet okay, if not communicated/posted during test; no communication with others, including sites such as, but not limited to Chegg)

Examiner(s): SD Ramsay

*Note: as this is a formal assessment, late submissions will not be accepted.
This means that if you do not submit your responses successfully by the
deadline you will not be able to submit.*

1. Modern, high-efficiency furnaces are known as condensing furnaces, meaning that the water vapour produced from combustion of the fuel is condensed to reclaim the enthalpy of vaporization to contribute to heating the air. Assuming 3582 g of water vapour per hour in the furnace exhaust gases at 218°C is condensed and cooled to 26°C, how much heating power is contributed through the condensation process? The molar mass of water is 18.015 g/mol, the enthalpy of fusion is 6.01 kJ/mol, the enthalpy of vaporization is 40.7 kJ/mol, and the molar heat capacity of the gas, liquid, and solid are 36.4 J/mol·K, 75.3 J/mol·K, and 37.7 J/mol·K, respectively. Note that 1 W = 1 J/s.

- (a) 1.69 kW
- (b) 2.8 kW
- (c) 0.56 kW
- (d) 2.64 kW

Correct answers: (b)

2. Which of the following strengthening mechanisms would you expect to change the Young's modulus of a metal?

- (a) None of these.
- (b) Grain size reduction.
- (c) Cold working.
- (d) Minor alloying additions (solid solution).

Correct answers: (a)

3. A humidifier is used to add moisture to the air in a room to make it more comfortable. A steam humidifier heats water until it boils to humidify the air. How much power is required for the humidifier to take 1883 g of water per hour at 19°C and boil all of it to vapour? The molar mass of water is 18.015 g/mol, the enthalpy of fusion is 6.01 kJ/mol, the enthalpy of vaporization is 40.7 kJ/mol, and the molar heat capacity of the gas, liquid, and solid are 36.4 J/mol·K, 75.3 J/mol·K, and 37.7 J/mol·K, respectively. Note that 1 W = 1 J/s.

- (a) 1223 W
- (b) 1359 W
- (c) 1005 W
- (d) 178 W

Correct answers: (b)

4. An unknown polymer sample is heated and observed to not melt before burning. Which of the following are possible microstructures for this polymer?
- A. Highly branched
 - B. Highly cross-linked
 - C. Linear
 - D. Network
- (a) B and C.
(b) B and D.
(c) B only.
(d) C and D.

Correct answers: (b)

5. An hypothetical polymer has the following molecular weight distribution. Determine the number average molecular weight of this polymer.

Mean Molecular Weight (g/mol)	Number Fraction
6300	0.14
23800	0.19
25100	0.49
36700	0.18

- (a) 34875.0 g/mol.
(b) 24309.0 g/mol.
(c) 104450.0 g/mol.
(d) 6077.2 g/mol.

Correct answers: (b)

6. An hypothetical ceramic has the cesium chloride crystal structure. The anion and cation radii are 0.172 nm and 0.138 nm, respectively. The anion and cation molar masses are $26.19 \frac{g}{mol}$ and $44.69 \frac{g}{mol}$, respectively. Which of the following is the theoretical density of this ceramic?

- (a) $2.57 \frac{g}{cm^3}$.
- (b) $10.26 \frac{g}{cm^3}$.
- (c) $20.53 \frac{g}{cm^3}$.
- (d) $82.12 \frac{g}{cm^3}$.

Correct answers: (a)

7. In the iron carbon system, the maximum solid solubility of carbon in ferrite occurs at 0.022 wt%C and 727°C, the eutectoid composition is 0.76 wt%C, the eutectoid temperature is 727°C and the cementite composition is 6.7 wt%C. For a 4232 g sample of 1.2 wt%C steel at 726°C, determine the mass of ferrite present.

- (a) 2354 g
- (b) 747 g
- (c) 3485 g
- (d) 1878 g

Correct answers: (c)

8. Covalently bonded materials are necessarily electrical insulators?

- (a) False.
- (b) True.

Correct answers: (a)

9. Introducing significant cross-links to a a sample of a hydrocarbon polymer would most likely cause which of the following changes?

- (a) Decrease in the yield strength and resistance to chemical dissolution
- (b) Decrease in the elastic constant.
- (c) No changes would be expected.
- (d) Increase in yield strength and resistance to chemical dissolution.

Correct answers: (d)

10. The intrinsic carrier concentration of a hypothetical compound semiconductor at room temperature is $1.70 \times 10^{12} \frac{1}{m^3}$ and the electron and hole mobilities are $0.7 \frac{m^2}{V \cdot s}$ and $0.18 \frac{m^2}{V \cdot s}$, respectively. Determine the intrinsic conductivity of this semiconductor at room temperature.

- (a) $1.50 \times 10^{12} \frac{1}{\Omega \cdot m}$
- (b) $1.91 \times 10^{-7} \frac{1}{\Omega \cdot m}$
- (c) $2.40 \times 10^{-7} \frac{1}{\Omega \cdot m}$
- (d) $4.91 \times 10^{30} \frac{1}{\Omega \cdot m}$

Correct answers: (c)

11. An hypothetical ceramic has the zinc sulfide crystal structure. The anion and cation radii are 0.19 nm and 0.046 nm, respectively. The anion and cation molar masses are $25.06 \frac{g}{mol}$ and $43.3 \frac{g}{mol}$, respectively. Which of the following is the theoretical density of this ceramic?

- (a) $5.61 \frac{g}{cm^3}$.
- (b) $0.7 \frac{g}{cm^3}$.
- (c) $22.44 \frac{g}{cm^3}$.
- (d) $2.8 \frac{g}{cm^3}$.

Correct answers: (d)

12. Which of the following is the correct coordination number for atoms within the BCC crystal structure?

- (a) 12.
- (b) 6.
- (c) 8.
- (d) 4.

Correct answers: (c)

13. Which of the following is most likely to be correct when comparing the density for samples of polyethylene?
- (a) Highly crystalline $\approx$ low crystallinity, high molecular weight $\approx$ low molecular weight...
 - (b) Highly crystalline $\approx$ low crystallinity, high molecular weight $>$ low molecular weight.
 - (c) Highly crystalline $>$ low crystallinity, high molecular weight $>$ low molecular weight.
 - (d) Highly crystalline $>$ low crystallinity, high molecular weight $\approx$ low molecular weight.

Correct answers: (d)

14. The maximum solid solubility of carbon in ferrite occurs at 0.022 wt% C and 727 ° C, the eutectoid composition is 0.76 wt% C, the maximum solid solubility of carbon in austenite occurs at 2.14 wt% C and 1147 ° C, and the cementite composition is 6.7 wt% C. For 2122 g of a 3.2134 wt% C steel at 726 ° C, what will be the mass of carbon present as part of a compound?
- (a) 67.9 g
 - (b) 74.2 g
 - (c) 58.7 g
 - (d) 83.5 g

Correct answers: (a)

15. A non-metallic material has a bandgap of 2.1 eV. In terms of the wavelengths of light that passes through the material, which of the following is true?
- (a) The material passes only light with wavelengths shorter than 532 nm.
 - (b) The material passes only light with wavelengths between 532 nm and 650 nm
 - (c) The material passes only light with wavelengths longer than 591 nm.
 - (d) The material passes all light except that with wavelengths between 532 nm and 650 nm

Correct answers: (c)

16. An hypothetical FCC metal has a density of $7.4 \frac{g}{cm^3}$ and a molar mass of $55.3 \frac{g}{mol}$. Which of the following is the correct number of atom sites (that is, without any vacancies)?

- (a) $1.09e+22 \frac{atoms}{cm^3}$
- (b) $1.34e-01 \frac{atoms}{cm^3}$
- (c) $6.80e-22 \frac{atoms}{cm^3}$
- (d) $8.06e+22 \frac{atoms}{cm^3}$

Correct answers: (d)

17. You are to choose the best material for use as a load bearing hinge surface. Without knowing anything else, how would you rank the following polymers from most appropriate to least appropriate for producing these moulds?

- A. Low molecular weight highly branched polyethylene
- B. High molecular weight and lightly branched polyethylene
- C. Ultrahigh molecular weight and highly branched polyethylene
- D. Ultrahigh molecular weight and lightly branched polyethylene

- (a) D, C, B, A
- (b) B, C, D, A
- (c) D, B, C, A
- (d) B, C, A, D

Correct answers: (a)

18. An hypothetical polymer has the following molecular weight distribution. Estimate the weight average molecular weight of this polymer.

Mean Molecular Weight (g/mol)	Number Fraction
7300	0.05
23200	0.17
28000	0.48
48200	0.3

- (a) 36246.3 g/mol.
- (b) 11477.9 g/mol.
- (c) 32209.0 g/mol.
- (d) 12587.0 g/mol.

Correct answers: (a)

19. A hypothetical FCC metal has a density of $7.49 \frac{g}{cm^3}$, a molar mass of $43.52 \frac{g}{mol}$ and an energy for vacancy formation of $0.74 \frac{eV}{atom}$. The Boltzmann constant is $8.617 \times 10^{-5} \frac{eV}{K}$. Which of the following is the correct number of vacancies in this metal at equilibrium at $859^\circ C$?

- (a) $2.29e+21 \frac{vacancies}{cm^3}$
- (b) $5.27e+19 \frac{vacancies}{cm^3}$
- (c) $2.04e+26 \frac{vacancies}{cm^3}$
- (d) $4.73e+18 \frac{vacancies}{cm^3}$

Correct answers: (b)

20. Copper and nickel are completely soluble in one another. This can best be explained by which of the following points?

- A. Cu and Ni have similar atomic radii.
- B. Cu and Ni have similar electronegativity.
- C. Cu and Ni have the same crystal structure.
- D. Cu and Ni have nearly identical melting temperatures.

- (a) A, B and C.
- (b) A, B and D.
- (c) A, C and D.
- (d) B, C and D.

Correct answers: (a)

21. An hypothetical ceramic has the sodium chloride crystal structure. The anion and cation radii are 0.168 nm and 0.102 nm, respectively. The anion and cation molar masses are $21.68 \frac{g}{mol}$ and $46.32 \frac{g}{mol}$, respectively. Which of the following is the theoretical density of this ceramic?

- (a) $2.87 \frac{g}{cm^3}$.
- (b) $1.43 \frac{g}{cm^3}$.
- (c) $0.72 \frac{g}{cm^3}$.
- (d) $5.74 \frac{g}{cm^3}$.

Correct answers: (a)

22. Two 1 m lengths of a woven polypropylene rope having a glass transition temperature of -20°C are loaded in tension for 8 months. Sample 1 is loaded at 25°C and sample 2 is loaded at 90°C . A third sample, 3 is left unloaded, while a fourth sample, 4 is loaded in tension at -45°C . Comparing the lengths of these rope samples at the end of this 8 month period, which of the following are possible?
- A. $1 > 2 > (3 \approx 4)$
 B. $2 > 1 > (3 \approx 4)$
 C. $2 > 1 > (3 > 4)$
 D. $1 > 2 > (4 > 3)$
- (a) A and C.
 (b) B only.
 (c) C and B.
 (d) A only.

Correct answers: (b)

23. Which of the following is the correct electronic configuration for a neutral atom of vanadium (V), in its ground (unexcited) state?
- (a) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^3$.
 (b) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$.
 (c) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$.
 (d) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$.

Correct answers: (b)

24. An hypothetical element has the diamond cubic crystal structure. The atomic radius is 0.19 nm and the molar mass is $27.57 \frac{\text{g}}{\text{mol}}$. Which of the following is the theoretical density of this element?
- (a) $232.53 \frac{\text{g}}{\text{cm}^3}$.
 (b) $0.54 \frac{\text{g}}{\text{cm}^3}$.
 (c) $0.27 \frac{\text{g}}{\text{cm}^3}$.
 (d) $4.34 \frac{\text{g}}{\text{cm}^3}$.

Correct answers: (b)

25. An hypothetical ceramic has a crystal structure in which only half of the interstitial sites are occupied and the cations occupy simple cubic interstitial sites. The anion and cation radii are 0.18 nm and 0.145 nm, respectively. The anion and cation molar masses are $28.57 \frac{g}{mol}$ and $34.4 \frac{g}{mol}$, respectively. Which of the following is the theoretical density of this ceramic?

- (a) $1.98 \frac{g}{cm^3}$.
- (b) $7.91 \frac{g}{cm^3}$.
- (c) $0.25 \frac{g}{cm^3}$.
- (d) $0.99 \frac{g}{cm^3}$.

Correct answers: (d)

26. A humidifier is used to add moisture to the air in a room to make it more comfortable. Humidifiers are most often used in the winter, when the house is being heated. An evaporative humidifier passes air over a wet filter and allows the water to evaporate as the air passes through the filter. Assuming the water vapour does not condense within the house, how much additional power is required to heat a home if 2195 g of water per hour at 21°C is evaporated into air that is heated to 24°? The molar mass of water is 18.015 g/mol, the enthalpy of fusion is 6.01 kJ/mol, the enthalpy of vaporization at this temperature is 44 kJ/mol, and the molar heat capacity of the gas, liquid, and solid are 36.4 J/mol·K, 75.3 J/mol·K, and 37.7 J/mol·K, respectively. Note that 1 W = 1 J/s.

- (a) 1493 W
- (b) 5 W
- (c) 1485 W
- (d) 95 W

Correct answers: (a)

27. Intramolecular bonds within a sample of a hydrocarbon polymer are which of the following types of bonds?

- (a) Ionic.
- (b) Covalent.
- (c) van der Waals.
- (d) Metallic.

Correct answers: (b)

28. Which of the following is the best description of the basis for metallic strengthening mechanisms?
- (a) Crystalline imperfections are removed thereby inhibiting the movement of linear crystalline imperfections.
 - (b) Crystalline imperfections are introduced that inhibit the movement of linear crystalline imperfections.
 - (c) Flaws at the surface of materials as well as internal pores are reduced thereby reducing the likelihood of a material cracking.
 - (d) The density of linear crystalline imperfections are reduced thereby inhibiting the movement of other linear crystalline imperfections.

Correct answers: (b)

29. In the iron carbon system, the maximum solid solubility of carbon in ferrite occurs at 0.022 wt%C and 727°C, the eutectoid composition is 0.76 wt%C, the eutectoid temperature is 727°C and the cementite composition is 6.7 wt%C. For a 652 g sample of 0.5118 wt%C steel at 728°C, determine the mass of austenite present.
- (a) 151 g
 - (b) 433 g
 - (c) 219 g
 - (d) 501 g

Correct answers: (b)

30. Which of the following is the best explanation for the continued load bearing following necking that can be observed in some thermoplastic polymers?
- (a) Chain orientation within the locally deformed region increases the polymer strength beyond the strength of the regions adjacent to the neck.
 - (b) Localized deformation in the neck increases the defect population, specifically vacancies, which limit dislocation movement and strengthen the polymer within the neck significantly more than the strength of the regions adjacent to the neck.
 - (c) Plastic deformation results in broken covalent bonds which then re-form as cross-links thereby increasing the polymer strength beyond the strength of the regions adjacent to the neck.
 - (d) Grain refinement within the necked region increases the strength through grain size reduction significantly more than the strength of the regions adjacent to the neck.

Correct answers: (a)